

Pirá 2.0 Dataset Analysis for Multiple-Choice Question Validation

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1. Dataset Description

[The Pirá dataset](#) is a bilingual (Portuguese-English) question-answering (QA) dataset focused on topics related to the ocean, the Brazilian coast, and climate change. The data has been pre-split into a train, test, and validation set on the GitHub repository. The size of the train dataset is 1806 rows $\times$ 31 columns. The size of the test dataset is 227 rows $\times$ 22 columns. The size of the validation dataset is 225 rows $\times$ 22 columns. Each row of the dataset contains a question, its answer, and context (like supporting text, abstract or excerpt) of it. The list of all columns are provided in the notebook for this milestone.

2. Data Analysis

Our initial EDA focused on obtaining a foundational understanding of the dataset. We reviewed key columns such as `question_en_origin` (the original English question), `at_labels` (which indicates whether a question is answerable), and `corpus` (which identifies the source of the passage as either a UN report or scientific abstract). One column in particular caught our attention: the abstract column.

The abstract field provides the supporting passage used to answer the corresponding multiple-choice question. These passages are either drawn from UN reports or scientific articles and represent the core context from which each question is derived.

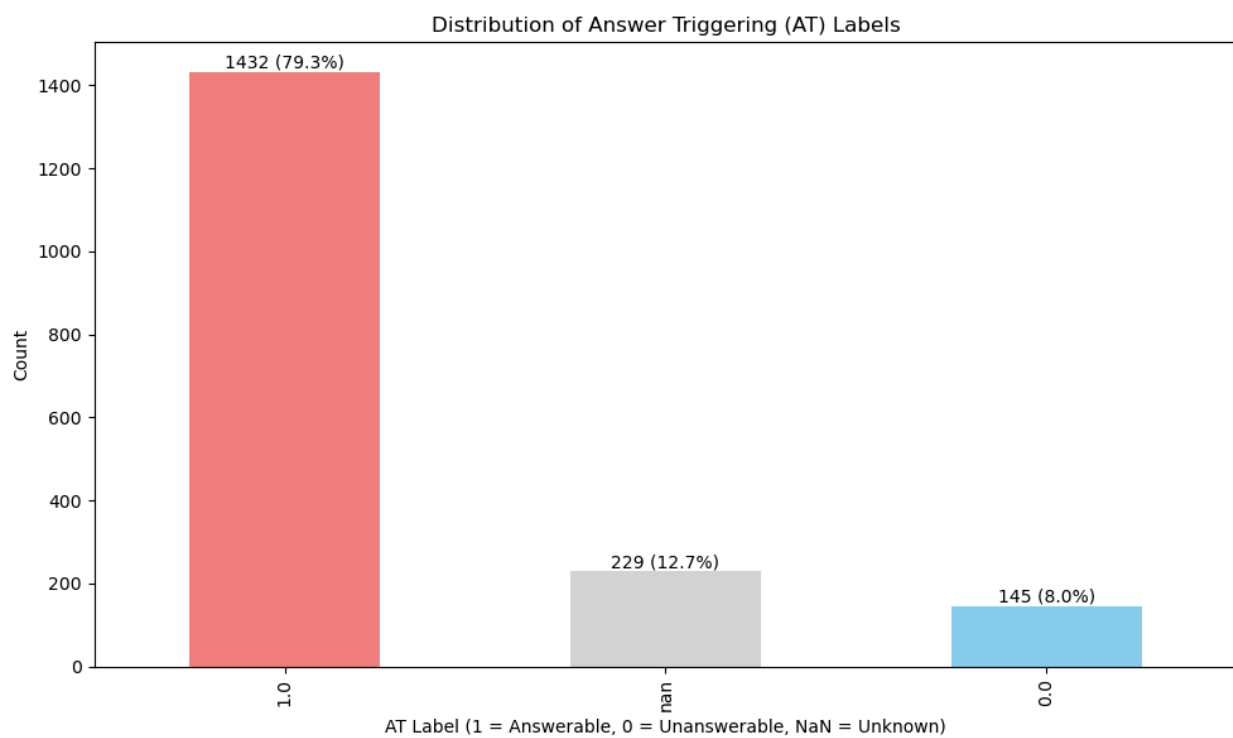
In this section, we explored passage-level patterns, especially those related to reuse frequency and question linkage.

- We analyzed how often each unique abstract appears in the dataset. While most passages were used only once or twice, we found that some are linked to as many as 17 different questions.
- These frequently reused passages were predominantly scientific abstracts, which are information-dense and cover a broader range of content than UN excerpts.
- We also printed the full text of these high-usage abstracts alongside all associated questions. This helped us uncover clusters of questions that share the same passage.

3. Meaningful Insights / Summary of Findings

We are interested in 2 columns: `answerability` and `answer_in_text`. Per Pira: “Answer Triggering is the problem of finding which questions should be answered and which ones should not; the

“answerability” label is created from the manual assessments for question meaningfulness.” Since each row cites a text (either from corpus 1 or 2), the `answer\_in\_text` column refers to whether that question and answer pair could be found directly in the text.



The dataset displays a significant class imbalance: among QA pairs with valid answerability labels, 79% are marked *answerable* and only 8% unanswerable. The remaining 13 % have missing *at_labels* and were excluded from this analysis. A similar skew appears in *answer_in_text*: 95 % of pairs are found in the passage, 3 % are *not* found, and 2 % are missing that field.

This imbalance prompted us to sharpen our research question. MCQ validation asks whether the correct answer can be derived *solely* from the passage, implicitly presuming the question is answerable. Therefore, while such imbalance is crucial for answer-triggering tasks (already explored in Pirá 2.0), it is not the primary concern for our MCQ-validation objective. We will filter out all “unanswerable” or unlabeled items before training.

4. Revised Project Question

Given a bilingual (English-Portuguese) passage from Pirá and a multiple-choice question whose correct answer is known to appear in the passage, can the model auto validate a multiple choice question—whether it truly requires reading the passage to answer correctly?

5. Model Implementation Plan

We plan to have four steps to our model's implementation. First, to balance computational capacities with cross-lingual comprehension, we will use a compact multilingual transformer from HuggingFace, like *DistilBERT-multilingual-cased* or *XLNet-small* (~ 80 to 140 million parameters). We filter the Pira 2.0 corpus to retain answerable MCQs and format them into five separate passage, question, candidate-answer sequences.

Second, we will reframe the multiple-choice problem as a five-way classification – the five examiner-provided answers (e.g. A to E). By casting answerability and answer selection as one joint softmax over five labels, the model can determine in a single forward pass if the passage contains the answer and, if so, identify the correct option. Third, we would employ the *MultipleChoice* variant of the selected transformer, which shares on an encoder for all six inputs and then applies a lightweight linear head to the pooled representations, yielding six logit-form outputs. We would then fine-tune the model using initial parameters detailed in the notebook.

Third, we will evaluate the model. The most intuitive measure is the fraction of examples where the predicted label matches the true label. To measure how good the model is at answering answerable questions, we will just compute the MCQ accuracy on the subset of answerable questions. We could further measure this retrieval task using Mean Reciprocal Rank, capturing how highly the system ranks the correct answer among all other candidates. This effectively simulates how confident our model is in its answer. Fourth, in our testing phase, each new question-passage pair is tokenized into six candidate sequences, passed into our fine-tuned model, and the highest-scoring index is extracted. The model will answer the question, which we will compare to the true labels.

Finally, this will give us a model that can predict the correct choice. For MCQs validation, we then run a second pass with the passage removed. If the model's top-choice changes or its soft-max confidence drops, we mark the item valid (answer truly passage-dependent); invalid otherwise.